

SOFTWARE ENGINEERING

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Semester & Section :- 5th Semester , B Section

Course Code:- UE24CS341A

Topic:- Lab 4 - Vibe Coding - Fishing Game

Deliverables to be done:-

1] Before Video Link:-

<https://drive.google.com/file/d/1RRLxCrnX0q4k417KCz6MXJaaW1zmEFg3/view?usp=sharing>

Before Video: Demonstrates the original Fishing Lab behavior before applying the Vibe Coding changes.

2] After Video Link:-

<https://drive.google.com/file/d/1QPWcm15dJuJbOxDI0STedPkWD41YihEz/view?usp=sharing>

After Video: Demonstrates the final Fishing Lab after implementing Tasks 1–4 and fixing the intermediate fish-visibility issue.

3] Prompt Evidence Link/ Chat History [Public Link]

<https://chatgpt.com/share/6ac6a1a1-c7a4-83ec-a3bf-9822e4c7fa0b>

Prompt 1 – Fix Fish Catch Detection Bug

Prompt 2 – Add Multiple Fish Types

Prompt 3 – Implement Player-Controlled Casting

Prompt 4 – Fix Fish Visibility and Implement 30-Second Round Timer

4] Features/Bug Fixed Snippet

Task 1 – Fish Catch Detection

Bug: Fish were being caught based only on their vertical/depth position, even when the hook was not horizontally aligned with the fish.

Fixed Snippet:

```
hook_rect = hook.get_rect()
```

```
for fish in fish_list:
    if hook_rect.colliderect(fish.get_rect()):
        return fish
```

The catch detection now requires actual overlap between the hook and fish.

Commit: [\[Task 1 commit\]](#)

Task 2 – Multiple Fish Types

Feature: Added multiple fish types with different movement speeds, sizes, colors, and point values.

Snippet:

```
class SlowFish(Fish):
    def __init__(self, x, y):
        super().__init__(x, y, speed=1.5, width=32, height=16, point_value=10, color=(50, 150, 255))

class FastFish(Fish):
    def __init__(self, x, y):
        super().__init__(x, y, speed=-4, width=44, height=22, point_value=25, color=(255, 165, 50))
```

Different fish now award different points when caught.

Commit: [792949c](#)

Task 3 – Player-Controlled Casting

Feature: Changed the hook from automatic casting to player-controlled casting using the Spacebar.

Snippet:

```
def start_cast(self):
    if self.hook.state == IDLE:
        self.hook.start_cast()
```

The Spacebar starts a cast only when the hook is idle. An active cast cannot be interrupted by another cast.

Commit: [\[Task 3 commit\]](#)

Task 4 – 30-Second Round Timer

Feature: Added a 30-second countdown timer, round-ending behavior, final score display, and the ability to start a new round.

Snippet:

```
ROUND_DURATION = 30
```

```
self.round_start_time = time.time()
self.round_over = False
```

The remaining time is displayed during the game. When the timer reaches zero, the round ends, further catches are disabled, and the final score is displayed.

A new round can be started using:

```
elif event.key == pygame.K_r and engine.round_over:
    engine.start_new_round()
```

The new round resets the score, timer, hook, and fish.

Commit: [6974aa8](#)

5. Final Git Commit History

6974aa8 Complete Task 4: add round timer and final game fixes

[Task 3 commit] Implement Task 3: player-controlled casting

792949c Implement Task 2: multiple fish types

[Task 1 commit]